

Swaroop Mudholkar

Electronics & Telecommunication Engineering Student | Embedded Systems | Automotive Electronics | GenAI | Computer Vision

✉ swaroopmudholkar@gmail.com ☎ +91 8830157488 📍 Pune, India

🌐 [linkedin.com/in/swaroop-mudholkar-entc6031a](https://www.linkedin.com/in/swaroop-mudholkar-entc6031a)

🔗 swaroop-mudholkar-portfolio.vercel.app 🐙 github.com/swaroop6031



EDUCATION

B.Tech in Electronics & Telecommunication <i>Pimpri Chinchwad College of Engineering</i>	2023 – 2027 Pune
12th Science, Maharashtra Board <i>S. B. Patil College of Science and Commerce</i>	2023 Pune
10th, CBSE <i>S. B. Patil Public School</i>	2021 Pune

SKILLS

- **Programming & Scripting:** Embedded C, C, C++, Python, CAPL, HTML/CSS, JavaScript
- **Embedded Systems, Automotive & AI:** STM32, ESP32, CAN Bus, Vector CANoe, CANdb++, Diagnostic Protocols (DTC), Sensor Interfacing (UART, I2C), YOLOv7/v8, OpenCV
- **Developer & Hardware Tools:** STM32CubeIDE, E-Plan Electric P8, MATLAB, Git, Multisim, ST-LINK

CERTIFICATES

Microcontroller Embedded C Programming: Absolute Beginners: Udemy (FastBit Embedded Brain Academy) Aug 2026

PROFESSIONAL EXPERIENCE

Cotmac Electronics Pvt. Ltd. June 2025 – July 2025 | Pune
PLC Panel Design and Testing Intern

- Engineered PLC panel schematics in E-Plan Electric P8, ensuring 100% wiring accuracy for industrial automation.
- Executed hardware-in-the-loop (HIL) signal testing and fault diagnostics across industrial control panels.

PCCOE EnTC Internship Cell, Pune
Documentation Team Lead

- Directed a student team managing technical documentation, reporting, and candidate tracking for 100+ students.
- Coordinated with departmental faculty and industry leads to streamline internship drives and placement workflows.

Technical Competitions

- Participated in Smart India Hackathon (SIH) 2024, IETE National Level Project Competition, and E-Yantra 2025.

PROJECTS

Proactive Detection of Aspergillus Growth Through Indoor Air Monitoring
Tech: STM32, UART, DHT22, MQ135, LCD

- Architected an STM32-based air-quality monitoring system using UART communication and multi-sensor telemetry to process environmental data streams in real time.
- Authored and presented peer-reviewed research on early fungal spore detection, published in the Scitepress International ICITSC Conference proceedings (2025).

Smart Chair System for Posture Monitoring & Wellness Reminder
Tech: ESP32, Force Resistor Sensors, Ultrasonic & Touch Sensors

- Built an ESP32 edge-monitoring system using multi-sensor fusion to process posture data and trigger feedback.
- Calibrated force and ultrasonic sensor thresholds to accurately distinguish between proper seating and poor posture in real time.

Engine ECU Simulation & CAN Network Validation
Tech: Vector CANoe, CAPL, CANdb++

- Simulated high-speed CAN networks in Vector CANoe with custom .dbc databases for multi-ECU validation.
- Developed CAPL scripts to automate Diagnostic Trouble Code (DTC) fault injection and live signal analysis.

Real-Time Object Detection using YOLOv7 & YOLOv8
Tech: Python, CNN, OpenCV, PyTorch

- Benchmarked YOLOv7 vs YOLOv8 models on PyTorch to optimize inference speed and accuracy for edge devices.
- Processed and annotated custom datasets, evaluating mAP, FPS, and detection accuracy across real-time video feeds.